

1. Configure, Verify, and Troubleshoot Port Security

What is Port Security?

Port security is a Layer 2 security feature on switches that limits access to a port based on MAC addresses. It helps prevent unauthorized devices from connecting to the network.

Key Features:

- Limits number of MAC addresses on a port.
- Defines which MAC addresses can access the port.
- Violations can be configured to take action: shutdown, restrict, or protect.
- Can be configured dynamically or statically.

Basic Configuration Example (Cisco IOS):

bash

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```
Switch(config)# interface FastEthernet0/1 Switch(config-if)# switchport mode access Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2 Switch(config-if)# switchport port-security violation restrict Switch(config-if)#
switchport port-security mac-address sticky
```

- maximum 2: Allows 2 MAC addresses on the port.
- violation restrict: Restricts and logs violation but doesn't shutdown port.
- mac-address sticky: Learns and saves MAC addresses dynamically.

Verification:

bash

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Switch# show port-security interface FastEthernet0/1 Switch# show port-security

Troubleshooting:

- Check violation counts.
- Check if the port is shut down due to violation.

- Verify configured MAC addresses.
- Use logs to find violation reasons.

2. Common Access Layer Threat Mitigation

- **Port Security:** Limits MAC addresses on switch ports.
- **BPDU Guard:** Protects STP topology by disabling ports receiving unexpected BPDUs.
- **DHCP Snooping:** Prevents rogue DHCP servers.
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Dynamic ARP Inspection (DAI): Prevents ARP spoofing attacks.

- IP Source Guard: Blocks packets from unauthorized IP/MAC addresses.
- Storm Control: Limits broadcast/multicast/flood traffic to prevent DoS.

3. Configure, Verify, and Troubleshoot IPv4 and IPv6 Access Lists (ACLs)

Access Control Lists (ACLs)

ACLs filter traffic to allow or deny based on criteria like IP address, protocol, port, etc.

IPv4 ACL Example (Standard):

bash

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```
Switch(config)# access-list 10 permit 192.168.1.0 0.0.0.255 Switch(config)# interface FastEthernet0/1 Switch(config-if)# ip access-group 10 in
```

IPv4 ACL Example (Extended):

bash

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```
Switch(config)# access-list 100 permit tcp any any eq 80 Switch(config)# interface FastEthernet0/1 Switch(config-if)# ip access-group 100 in
```

IPv6 ACL Example:

bash

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```
Switch(config)# ipv6 access-list BLOCK-HTTP Switch(config-ipv6-acl)# deny tcp any any eq 80 Switch(config-ipv6-acl)# permit ipv6 any any
Switch(config)# interface GigabitEthernet0/1 Switch(config-if)# ipv6 traffic-filter BLOCK-HTTP in
```

Verification:

bash

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```
Switch# show access-lists Switch# show ip interface FastEthernet0/1 Switch# show ipv6 access-list
```

Troubleshooting:

- Verify ACL is applied on correct interface and direction.
- Check ACL rules order (first match applies).
- Use counters to check matches (show access-lists).

4. Verify ACLs Using APIC-EM Path Trace ACL Analysis Tool

Cisco APIC-EM Path Trace helps visualize the path of traffic and analyze ACLs affecting the flow.

- It shows if ACLs permit or deny traffic.

- Useful for troubleshooting ACLs in complex environments.
 - Requires Cisco APIC-EM controller setup and devices managed via it.
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5. Configure, Verify, and Troubleshoot Basic Device Hardening

Device Hardening Steps:

- Change default passwords.
- Disable unused services and ports.
- Use secure management protocols (SSH, HTTPS).

- Set login banners.
- Limit access via ACLs.
- Enable logging and monitor events.
- Apply port security.
- Use SNMPv3 for secure monitoring.

Example:

bash

```
Switch(config)# no ip http server Switch(config)# service password-encryption Switch(config)# enable secret StrongPassword Switch(config)#  
line vty 0 4 Switch(config-line)# transport input ssh Switch(config-line)# login local
```

6. Device Security Using AAA with TACACS+ and RADIUS

What is AAA?

AAA stands for Authentication, Authorization, and Accounting — critical for device access control and logging.

TACACS+ vs RADIUS:

Feature	TACACS+	RADIUS
Protocol	TCP	UDP

Encrypts entire packet	Yes	Only password encrypted
Use	Device management	Network access
Granular command control	Yes	No

Basic TACACS+ Configuration:

bash

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```
Switch(config)# tacacs-server host 192.168.1.10 key MySecretKey Switch(config)# aaa new-model Switch(config)# aaa authentication login
default group tacacs+ local Switch(config)# aaa authorization exec default group tacacs+ local Switch(config)# line vty 0 4 Switch(config-line)#
login authentication default
```

Basic RADIUS Configuration:

bash

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```
Switch(config)# radius-server host 192.168.1.20 key MySecretKey Switch(config)# aaa new-model Switch(config)# aaa authentication login  
default group radius local Switch(config)# line vty 0 4 Switch(config-line)# login authentication default
```

Verification:

bash

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```
Switch# show aaa sessions Switch# show tacacs Switch# show radius
```